

Automated Essay Scoring: A Reflection on the State of the Art

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Abstract

While steady progress is made on the task of automated essay scoring (AES), much of the recent work in this area has focused on developing models that beat existing models on a standard evaluation dataset. While improving performance numbers remains an important goal in the short term, such a focus is not necessarily beneficial for the long-term development of the field. We reflect on the state of the art in AES research, discussing issues that we believe can encourage researchers to think bigger than improving performance numbers with the ultimate goal of triggering discussion among AES researchers on how we should move forward.

1 Introduction

Automated Essay Scoring (AES), the task of automatically assigning a *holistic* score to an essay that summarizes its overall quality, is arguably one of the most important applications in natural language processing (NLP). Given the large number of essays written by students from all over the world in both test and classroom settings, being able to automatically score essays could save a tremendous amount of manual grading effort.

Despite the fact that AES has been investigated for more than 50 years (Page, 1967), the task is still far from being solved. Nevertheless, AES has been progressing steadily. In recent years, a standard recipe for publishing in this area of research seems to involve proposing a sophisticated neural model that can beat competing models on a standard evaluation dataset such as ASAP (see Section 2.1 for details). While improving performance numbers remains an important short-term goal for AES research, such a focus is not necessarily beneficial for the long-term development of the field.

Our goal in this paper is to reflect on the state of the art in AES research. Specifically, we discuss issues that we believe can encourage researchers, particularly junior researchers, to think bigger than

merely improving performance numbers, with the ultimate goal of triggering discussion among AES researchers on how we should move forward.¹

2 Current State of AES Research

To facilitate our discussion of the future directions of AES research, in this section, we provide an overview of the current state of AES research.²

2.1 Corpora

While a number of AES corpora have been developed, one of them stands out as being the most extensively used in AES research in the past five years or so, namely ASAP (and its extension ASAP++). Introduced as part of a 2012 Kaggle competition, the Automated Student Assessment Prize (ASAP) corpus has become a popular dataset for holistic scoring, especially given its vast collection of essays per prompt (up to 3,000 for some prompts). ASAP facilitates the development of high-performing, prompt-specific systems. ASAP++ (Mathias and Bhattacharyya, 2018) is an extension of ASAP where each essay is scored along multiple traits. Note that ASAP is composed of three types of essays (namely, narrative/descriptive essays, persuasive essays, and source-dependent essays). Not all traits are applicable to all essay types. For instance, Organization and Conventions are scored for narrative/descriptive essays and persuasive essays only.

Other English corpora that have been developed for and used in AES research include: (1) TOEFL (Blanchard et al., 2013), a corpus of essays from the TOEFL exam that is originally developed for

¹While Beigman Klebanov and Madnani’s (2020) theme paper focuses more broadly on the emergent uses of automated writing technologies, we focus specifically on research directions that we think are worth pursuing by AES researchers.

²For a comprehensive overview of AES research prior to 2022, we refer the reader to the books published by Shermis et al. (2010) and Beigman Klebanov and Madnani (2021), as well as a survey published by Ke and Ng (2019).

the Native Language Identification task where the proficiency label that comes with each essay (Low, Medium, or High) is used as its holistic “score” for training AES systems; (2) the Cambridge Learner Corpus-First Certificate in English exam (CLC-FCE) (Yannakoudakis et al., 2011), where each essay is scored holistically and annotated with the linguistic error types it contains; (3) the International Corpus of Learner English (ICLE), where a subset of essays has been scored along multiple dimensions of essay quality, such as Organization (Persing et al., 2010) and Argument Persuasiveness (Persing and Ng, 2015); and (4) the Argument Annotated Essays (AAE) corpus (Stab and Gurevych, 2014), where each persuasive essay is scored based on the strength of its thesis (Ke et al., 2019) and the persuasiveness of its argument (Ke et al., 2018).

AES corpora in other languages exist, such as Ostling’s (2013) Swedish corpus, Horbach et al.’s (2017) German corpus, Marinho et al.’s (2021) Portuguese corpus, the GoodWriting dataset³ (in Japanese), and the MERLIN dataset⁴, which is composed of German, Italian, and Czech essays.

2.2 Evaluation Metric

The standard metric used to evaluate AES models is Quadratic weighted Kappa (QWK)⁵. QWK is an *agreement* metric that ranges from 0 to 1 but can be negative if there is less agreement than what is expected by chance. More specifically, QWK is a weighted version of Kappa where each case of disagreement (i.e., the (rounded) predicted score is different from the reference score) is weighted by the squared difference between the reference score and the predicted score. This allows the metric to distinguish between near misses and far misses.

2.3 Systems

Existing AES systems can be divided into three categories, as described below.

2.3.1 Heuristic Approaches

Virtually all early AES systems are *heuristic-based* and typically possess the following characteristics (e.g., Elliot (2003), Attali and Burstein (2006)):

Trait-driven holistic scoring. Many factors play a role when human raters score an essay holistically. These factors, which are known as essay

traits, correspond to different dimensions of essay quality, such as Organization, Coherence, Technical Quality (i.e., fluency, grammar, and mechanics), and Argument Persuasiveness. Motivated by the human essay scoring process, the holistic score returned by a heuristic AES system is typically computed as the weighted sum of the trait scores.

Heuristic trait-specific scoring. Given the lack of annotated data, each trait-specific score is computed using heuristics. For example, to compute the Organization score, which reflects how well-organized the essay is, the *e-rater* system (Attali and Burstein, 2006) determines whether the essay is organized as a 5-paragraph essay where the first paragraph is the introduction, the last paragraph is the conclusion, and the middle three paragraphs each presents a key point with supporting evidence. The functional role of each paragraph (e.g., Introduction) is determined heuristically.

Focus on non-content-based traits. Traits can broadly be divided into two categories: *content-based* traits, which are based on the essay’s content (e.g., Argument Persuasiveness, Coherence), and *non-content-based* traits, which are based on the surface realization of the content (e.g., Grammar, Fluency). Generally, the content-based traits are much harder to score than the non-content-based traits. For example, while Fluency and Grammaticality can be determined fairly easily using a language model and a grammar checker respectively, determining Argument Persuasiveness may require a deep understanding of the content. Content-based traits are particularly difficult to compute in the absence of labeled data. Consequently, heuristic approaches have largely focused on employing non-content-based traits for holistic scoring.

2.3.2 Machine Learning Approaches

As annotated AES corpora became publicly available, the focus of AES research also started to shift from heuristic approaches to machine learning approaches, where an off-the-shelf machine learning algorithm (e.g., SVM, linear regression) is used to train a classifier or a regressor for scoring (e.g., Briscoe et al. (2010)). AES research in the machine learning era has the following characteristics:

Focus on feature engineering. The focus is designing *low-level* and *high-level* features. Low-level features include *length-based* features (e.g., the number of tokens in the essay) (Yannakoudakis et al., 2011; Vajjala, 2018), *lexical* features (e.g., the presence/count of each n-gram) (Chen and

³<https://goodwriting.jp/wp/?lang=en>

⁴<https://www.merlin-platform.eu/>

⁵See <https://www.kaggle.com/competitions/asap-aes/overview/evaluation> for details.

He, 2013; Phandi et al., 2015), word embeddings (Cozma et al., 2018), *word category* features (e.g., whether a word is a modal) (Farra et al., 2015; McNamara et al., 2015), and *syntactic* features (e.g., part-of-speech tag sequences) (Chen and He, 2013). High-level features include *readability* features (i.e., metrics that reflect how easy it is to read the essay) (Zesch et al., 2015), *prompt-relevant* features (i.e., features that encode the similarity between the essay and the prompt it was written for) (Louis and Higgins, 2010; Klebanov et al., 2016), *argumentation* features (e.g., the number of claims in a persuasive essay) (Ghosh et al., 2016; Wachsmuth et al., 2016; Nguyen and Litman, 2018), *semantic* features (e.g., features derived from lexico-semantic resources such as FrameNet (Baker et al., 1998)) (Beigman Klebanov and Flor, 2013), and *discourse* features (e.g., local coherence features derived from Centering Theory (Grosz et al., 1995)) (Yannakoudakis and Briscoe, 2012).

Focus on within-prompt scoring. In within-prompt scoring, an AES model is trained on essays written for a prompt and then applied to test essays written for the *same* prompt. Some have argued that within-prompt scoring is not a practical setting. The reason is that when these within-prompt scorers are applied to essays written for a new prompt, their performance often deteriorates considerably. In practice, before they are applied to score essays written for a new prompt, they need to be retrained on scored essays written for the new prompt. However, manually scoring essays is time-consuming and requires a lot of expertise.

Learning-based trait-specific scoring. As machine learning approaches to AES became popular, researchers began to examine learning-based approaches to trait-specific scoring. The development of learning-based models for trait-specific scoring is facilitated by the release of annotated datasets where essays are scored along different essay traits (Persing and Ng, 2013, 2014, 2015). While the scoring of content-based traits is largely ignored in heuristic approaches, researchers have begun learning models for scoring content-based traits. Nevertheless, even with annotated data, the scoring of content-based traits remains a challenging task.

2.3.3 Deep Learning Approaches

With the advent of the neural NLP era, the vast majority of recently developed AES models are *deep learning-based*. AES research during this period can be summarized as (1) a focus on learning the

distributed (i.e., real-valued vector) *representation* of an essay (by adjusting the weights in a neural network) so that essays that are similar in quality will have similar representations and (2) an exploration of new, challenging AES task settings such as cross-prompt scoring and multi-trait scoring.

Early approaches. Early neural models combine CNNs and RNNs to capture spatial and temporal dependencies respectively. For instance, Taghipour and Ng (2016) first use a CNN to extract n-gram-level features to capture local dependencies and then use an LSTM to generate a long-distance representation of an essay for holistic scoring. These models are subsequently replaced by Transformer-based models, which possess a vast amount of linguistic and commonsense knowledge acquired from large, unlabeled corpora. For instance, Yang et al. (2020) proposed R²BERT, an AES model obtained by fine-tuning BERT. Wang et al. (2022) proposed a multi-scale BERT-based structure that captures (automatically learned) features at the token, segment, and essay levels. Uto et al. (2020) showed that neural AES models could be improved with hand-crafted features.

Neural AES models can be improved by exploiting document structure. Dong and Zhang (2016) viewed an essay as having a two-level hierarchical structure: an essay is composed of a sequence of sentences, each of which is composed of a sequence of words. Given this view, they designed a two-layer model where the first layer creates a representation for each sentence and the second layer creates an essay representation by combining sentence representations. Further improvements can be made via attention pooling (Dong et al., 2017).

Cross-prompt scoring. As noted above, some have argued that within-prompt scoring is not a practical setting for AES. Hence, researchers have recently begun working on the task of cross-prompt AES (Ridley et al., 2021), where the goal is to train a model that can offer good performance when it is applied to score essays written for unseen prompts.

A few approaches to this relatively new task of cross-prompt scoring have been developed. Cummins et al. (2016) recast cross-prompt scoring as a domain adaptation problem, where a prompt is viewed as a domain. Specifically, the goal is to use a domain adaptation method to adapt an AES model trained on source prompts to the target prompt. Do et al. (2023) incorporated as input for cross-prompt scoring essay prompt information,

which ironically is not exploited by many AES models. To facilitate generalization to new prompts, [Chen and Li \(2023\)](#) proposed a cross-prompt scoring model that seeks to make the source essay representations and the target essay representations more *consistent* with each other via contrastive learning, and several researchers have employed prompt-independent features in their AES models ([Jin et al., 2018](#); [Li et al., 2020](#); [Ridley et al., 2020](#)).

Multiple-trait scoring. Since the holistic score of an essay is influenced by its trait-specific scores, it would be natural to develop *joint* models for AES that simultaneously predict the holistic score and the various trait-specific scores. Multiple-trait scoring models can be obtained by several simple methods. The first method involves replacing the output layer of a holistic scoring model with multiple output layers, one for each trait ([Hussein et al., 2020](#)). The second method involves making multiple copies of a holistic scoring model where each copy is responsible for scoring one trait and the copies interact with each other via a shared representation layer ([Mathias and Bhattacharyya, 2020](#)). The third method is similar to the second, except that the predicted trait scores are used as input to predict the holistic score ([Kumar et al., 2022](#)).

Prompt-based approaches. [Mizumoto and Eguchi \(2023\)](#) investigated a prompt-based approach to AES. Here, the word “prompt” refers to the use of *prompting* in large language models (LLMs), not essay prompts. Prompt-based approaches are motivated by two key strengths of LLMs. First, LLMs possess a vast amount of commonsense knowledge that can be exploited to perform various tasks. Second, LLMs are good at understanding complex natural language instructions. Given these strengths, we can ask an LLM to perform a task as complex as AES by providing instructions in the form of a prompt that may include, for instance, the rubric in a zero-shot setting, where *no* manually scored essays are provided as training examples (e.g., [Lee et al. \(2024\)](#)), or a few-shot setting, where a few labeled examples are provided as part of the prompt (e.g., [Mansour et al. \(2024\)](#); [Xiao et al. \(2024\)](#)).

3 AES is more than improving QWK

What have we learned about AES over the years other than the fact that the QWK scores are improving on standard evaluation datasets? In the pre-neural NLP era when many AES researchers were

focusing on feature engineering, we could at least gain some insights into what kind of features would be useful for AES through feature ablation experiments. With neural AES models, the representation of an essay is learned. Many of them do not employ hand-crafted features, and the output cannot be interpreted easily (though some researchers have attempted to visualize using attention weights which words and/or phrases are considered important by the model). So it has become even harder to understand what has been improved.

Recommendation #1: We recommend that researchers understand the strengths and weaknesses of the systems they developed by performing a qualitative and quantitative analysis of the system outputs. For example, while it is not uncommon for AES researchers to report results that are better than existing results *when averaged over all essay prompts in the corpus*, without further analysis, a reader would not know where the improvements came from. Does the proposed model perform better on all types of essays (e.g., persuasive essays, narratives) or only certain types of essays, and if it performs better on all types of essays, are there certain essay types for which the improvement is more pronounced? Does the model perform better because it can better distinguish between essays that are similar w.r.t. particular traits (e.g., ORGANIZATION)? Is the model better because it scores the essays belonging to the minority classes better? Having answers to at least some of these questions could help identify areas of improvement. Furthermore, researchers have recently begun to explore the explainability of AES models ([He et al., 2022](#); [Fiacco et al., 2023](#)). Though research in this area is still in its infancy, a deeper understanding of how the decisions of a model are made could also help identify its strengths and weaknesses.

4 ASAP is not the only essay corpus

As mentioned before, in recent years many AES researchers evaluated their model solely on ASAP. A natural question is: have existing models overfitted ASAP? In other words, can models that perform well on ASAP be expected to generalize well to other essay corpora? Unfortunately, little analysis has been performed on ASAP that would enable us to gain a better understanding of whether certain corpus-specific characteristics are present in ASAP that could cause models to overfit the data. For example, what fraction of the persuasive essays in

ASAP made arguments that are persuasive? If the vast majority of these essays contained persuasive arguments, then a model trained on ASAP might not perform well on a corpus where it is important to distinguish between essays with persuasive arguments and those with unpersuasive arguments.

One piece of information that we know about ASAP is that the essays it contains were written by American students in Grades 7–10. While the native languages of these students are not known, the majority of them are expected to be native speakers of English. Given this context, ASAP is not an ideal corpus for the development and evaluation of AES systems. Recall that one of the primary motivations and key driving forces behind the development of AES systems is its application to scoring essays written by takers of standardized tests such as TOEFL and SAT, where many of the exam takers are learners of English as a second language (ESL). As is well-known to essay grading researchers, the common types of error made by ESL learners who speak different native languages are not the same (Yannakoudakis et al., 2011). Hence, it is possible that the ASAP essays are simpler to score because grammatical errors may be less of an issue in these essays. In contrast, for AES of essays written by non-native speakers, AES systems may need to be combined with grammatical error correction systems for more accurate modeling of the quality of an essay w.r.t. the Grammar trait.

Another piece of information that we know about ASAP (as well as other AES corpora where essays are written in a *time-restricted* setting such as a test setting) is that essay length (measured by the number of words) is often a confounding variable for holistic scoring. Consequently, it is not clear how models trained on ASAP would perform when applied to corpora where essays are written in a *time-unrestricted* setting, such as essays written as part of a homework with a length restriction (e.g., between 500 and 600 words), as length may no longer be a confounding variable in these corpora.

Recommendation #2: To understand whether models trained on ASAP can generalize well when applied to other essay corpora, we recommend that AES researchers evaluate their systems on not only ASAP but at least one other publicly available corpus, preferably a corpus where the native languages of the essay writers are different from the language in which the essays were written, such as CLC-FCE (Yannakoudakis et al., 2011) or a corpus where es-

says were written in a time-unrestricted setting, such as ICLE (Granger et al., 2009).

5 Is it time for cross-prompt AES?

Cross-prompt AES is a challenging task. Below we discuss the challenges from three perspectives.

Knowledge. For an AES model to perform well on a new prompt, we need knowledge specific to the new prompt. For example, if the new prompt is “write a persuasive essay on whether capital punishment should be abolished”, an AES model used to score essays for this prompt needs to distinguish between persuasive arguments and unpersuasive arguments for (or against) capital punishment. For cross-prompt scoring, this kind of knowledge needs to be extracted from external knowledge sources. One possibility is to prompt a large language model (LLM): a human first provides a few examples of persuasive and unpersuasive arguments for each stance, which are then used to elicit prompt-specific knowledge inherent in the LLM.

Training data. What kind of training data is needed for cross-prompt AES? If the test essays are persuasive essays, then ideally the training essays should also be persuasive essays. The reason is that the traits that affect the holistic score of a persuasive essay (e.g., Argument Persuasiveness) are not the same as those that affect the holistic score of a non-persuasive essay, even though there are traits that are common to all types of essays (e.g., organization). If the training set and the test set are composed of different types of essays, what is learned about good essays from the training set may not necessarily be applicable to good essays in the test set. Similarly for rubrics: the rubric used to score the training essays should be the same (or at least similar to) the one used to score the test essays; otherwise, knowledge of good essays that is learned from the training set may not be transferable to the test set because essays that are good according to the training rubric could be considered bad according to the test rubric.

However, this is not how Ridley et al.’s (2021) cross-prompt AES model was trained and evaluated. Specifically, they employed ASAP for training and evaluation. Recall that ASAP is composed of persuasive, narrative, and source-dependent essays written for eight prompts. Their cross-prompt AES experiments were conducted using leave-one-prompt-out cross validation, where they trained a model using essays for all but one prompt and

tested it on essays for the prompt that was left out. This means that the training set may be composed of essays that have a different essay type than that in the test set. Additionally, the training and test prompts often employ differing rubrics. It is not clear whether it even makes sense to train and evaluate the model using this setup.

Models. Can within-prompt AES models be used for cross-prompt AES? The vast majority of existing within-prompt AES models do *not* take the essay prompt and the scoring rubric as input. For within-prompt AES, this may be fine, as the essays in the training set and the test set are written for the same prompt and scored using the same rubric. For cross-prompt AES, this may not be fine, as (1) the model does not know what prompt-specific knowledge to extract from external sources without knowing the prompt and (2) the model does not know the criteria for good and bad essays without the rubric. Hence, it is not clear the extent to which the performance of Ridley et al.’s (2021) cross-prompt AES model is limited by the fact that it relies on neither the unseen prompt nor the rubric.

Recommendation #3: Cross-prompt AES is a challenging task that deserves the attention of AES researchers. We recommend that researchers focus on developing new annotated corpora, new methods for gathering prompt-specific knowledge, and new AES models for cross-prompt AES.

6 Traits are useful but under-explored

Two key challenges discussed above involve building AES models that are *interpretable* and *generalize* well to unseen prompts. Next, we propose to examine the use of traits to address both challenges.

Interpretability. If a holistic scoring model is built by using *only* traits as inputs (such as the heuristic AES models described in Section 2), then interpretability is relatively straightforward. For instance, if a linear model is used where the holistic score is computed as a weighted sum of the trait-specific scores, then the holistic score can be easily explained by the trait-specific scores. Even if a non-linear model is used, the attention weights can be used to shed light on how much influence a particular trait has on the holistic score. Not only do essay traits make interpretability straightforward, but the trait scores also serve to provide feedback to an essay’s writer. For example, if someone gets a low holistic score for her essay, she can check which trait-specific scores are low, and subsequently iden-

tify which aspects of her essay need improvement.

Cross-prompt generalizability. Recent effort on cross-prompt AES research has involved identifying prompt-independent features, such as length-based features. While traits have been used as features for holistic scoring, they have primarily been used for within-prompt rather than cross-prompt scoring. One reason, we believe, is that traits are rarely viewed as prompt-independent features.

Given the prompt-independent nature of traits, we propose to use them for cross-prompt AES. Specifically, we propose to view the set of traits commonly known to play a role in determining the holistic score of a particular type of essays (e.g., persuasive essays) as a prompt-independent *representation* of these essays. Note that this trait-based essay representation is dependent on essay type, as the traits that influence the holistic score of a persuasive essay are not necessarily the same as those that influence narrative essays, for instance.

In practice, however, using traits to address interpretability and cross-prompt generalizability is challenging. The key challenge stems from the fact that content-based traits are difficult to score accurately, even when labeled data is available for training content-based trait-scoring models. One possibility is leverage LLMs for scoring traits (including content-based traits) in a zero-shot setting, where we provide the rubric developed for a given trait as part of the instructions for an LLM when requesting it to score the trait.

Should we believe that LLMs can help us accurately score traits? As mentioned in Section 2, state-of-the-art LLMs are very good at understanding inputs as complex as essays. Since scoring content-based traits requires an understanding of the essay’s content, exploring the use of LLMs for trait scoring is a promising direction.

Nevertheless, it is worth noting that recent prompt-based approaches to *holistic* scoring (Lee et al., 2024) including those that employ a chain-of-thought approach (Xiao et al., 2024) are not as competitive as fine-tuned models in performance. While the underlying reasons are not yet clear, researchers have demonstrated that minor changes in the prompt Mansour et al. (2024), employing different decoding methods Shi et al. (2024), and variations in random seeds Dodge et al. (2020) can all result in significant performance changes.

Given these results, researchers should expect similar challenges when prompting LLMs for trait

scoring. However, while prompt-based approaches to holistic scoring seek to achieve performance as competitive as fine-tuned models, when using prompt-based approaches for trait scoring, the eventual goal is to use the resulting trait scores to train a holistic scoring model. In other words, even somewhat noisily computed trait scores could improve holistic scoring after fine-tuning.

Recommendation #3: We recommend that a more thorough investigation of the impact of traits on holistic scoring should be conducted as they could be a viable solution to key problems concerning neural model interpretability and cross-prompt model generalizability. Given the difficulty involved in computing content-based traits, we recommend that researchers examine how LLMs can be exploited, possibly via prompting-based approaches, to score content-based traits.

7 Corpus development is slow

The fact that ASAP is still the primary corpus used for evaluating AES models more than 10 years after its initial release is somewhat unusual in the NLP community for a task that is as popular as AES. This perhaps suggests that corpus development is seriously lagging behind model development. We believe the reasons are at least two-fold.

Assembling an essay corpus is by no means easy. For many other NLP tasks, assembling a large, unannotated corpus composed of news articles or tweets is relatively easy. This is not the case for essays: while we may be able to assemble a corpus of raw classroom essays, it may still require collaboration by multiple instructors over multiple years in order to obtain a sizeable corpus. Yet, even after multiple years of effort, the resulting corpus may still be too small to enable the pre-training of models specifically on raw essays.

There is no easy solution, however. Organizations that have access to a large number of essays written by people different demographic characteristics should consider releasing language resources to the AES community. These language resources would ideally be in the form of raw essays. We believe that this is doable: TOEFL11, for example, was released by Educational Testing Service (ETS) in 2013, and we encourage ETS and similar organizations to continue this effort. If essays cannot be released, then they may consider releasing models pre-trained on these essays. Language models pre-trained for specialized domains have been suc-

cessfully deployed (e.g., the legal domain (Xiao et al., 2021)). Unlike many other domains where it may suffice to have one pre-trained model per language, for AES research it may make sense to develop multiple pre-trained models that differ in terms of evaluation setting (e.g., whether the essays are written in a time-constrained or unconstrained setting), grade level (university undergraduates vs. high-school students, and native language.

Having models pre-trained on essays could enable them to acquire linguistic or even prompt-specific knowledge from raw essays. The hope is that with the availability of pre-trained language models for essays, a relatively small amount of labeled data will be needed to fine-tune them to perform specific essay-related tasks, such as AES.

Manually labeling essays is a labor-intensive and expensive procedure. The reason is that manual labeling of essays is typically performed by trained experts and cannot be reliably done via crowdsourcing, especially if it involves scoring along multiple traits. Even if we do have the time and resources to perform manual labeling of essays, the essay grading community has not developed a *vision* of what annotations would benefit the development of AES models as well as models for other essay-related tasks *in the long run*. Ideally, there would be one or two corpora that would contain multiple layers of annotation. For example, one layer would be composed of scores (i.e., the holistic and trait-specific scores), another layer would be composed of written feedback to essay writers, and a third layer would be composed of essays that have been revised by experts. Having multiple layers of annotation in the same corpus would be beneficial from the perspective of model development, as it would allow a joint model to be built that can, for instance, exploit the inter-dependencies among different layers. For example, scoring and the generation of written feedback can be modeled simultaneously, as positive feedback is typically associated with positive scoring. In other words, inter-layer constraints can be incorporated into the model architecture to capture such dependencies.

The layers that should be included in the annotation scheme should by no means be limited to the aforementioned possibilities: it is something that needs to be discussed by AES researchers and perhaps the wider essay grading community.

Recommendation #5: We recommend that the community devote more effort to corpus develop-

ment, as progress on model development will eventually be hindered by the arguably slow progress on corpus development. Research organizations such as ETS should consider releasing large corpora of raw, unannotated essays or language models that are pre-trained on such corpora. In addition, we encourage the AES community to develop a shared vision of what corpora, particularly annotated corpora, would benefit AES research in the long run. Ultimately, the community should discuss the possibility of creating a large, annotated corpus and the kind of manual annotations that are desirable to have in that corpus. This whole discussion could begin with a small, annotated essay corpus released as part of a shared task that contains some of the annotation layers described above. We believe that the shared task could bring together interested parties to discuss what the next steps should be as far as corpus development is concerned.

8 LLMs can be used to assist AES

So far, LLMs have been primarily used in prompt-based approaches to holistic scoring, with results that are less competitive than those of fine-tuned models. As researchers continue to improve prompt-based approaches, we encourage them to examine other ways of exploiting LLMs.

Rather than expecting LLMs to *replace* humans (as in prompt-based approaches to holistic scoring), we can instead aim to use LLMs to *assist* humans in AES-related tasks. In particular, it is worth investigating whether LLMs can be profitably used to assist corpus creation and annotation. As noted previously, assembling a large unlabeled essay corpus and performing multi-layer annotation of such a corpus are both daunting tasks. We believe LLMs can be used to perform the groundwork for these tasks. For example, given the difficulty in assembling an essay corpus, it is worth investigating whether it would be possible to ask LLMs to generate essays subject to a given set of constraints, such as essay length, grade level, and score level (given the corresponding score description taken from a rubric).⁶ As another example, since corpus annotation is a time-consuming process, we can ask LLMs to score traits given the corresponding rubrics (possibly by providing the rationale behind why the essay deserves a particular score), provide written feedback (given their strong generative ca-

pabilities), and revise essays in a zero-shot fashion.

While the LLM outputs can be far from perfect, if we aim to use LLMs to *assist* rather than *replace* humans, then the goal would be for humans to serve as *reviewers* of the LLM outputs. For example, for corpus creation, a human can examine the LLM-generated essays to determine if they conform to the given constraints. For corpus annotation, a human can review the LLM-produced trait scores, written feedback, or even the revised essay, with the goal of determining whether reviewing and correcting LLMs' outputs can help save a human annotator's time than performing these annotations from scratch. If so, then there is value in using LLMs in assisting in these tasks. For instance, human trait scoring could be accelerated by understanding the LLM's rationale behind assigning a trait score.

Recommendation #6: We recommend that AES researchers explore novel ways of exploiting LLMs, especially concerning how LLMs can be used to improve human efficiency in completing daunting tasks such as corpus creation and annotation.

9 Essay grading beyond scoring

While AES has largely been tackled as a standalone task, researchers can think about how AES systems can be combined with other technologies developed for AI in education. For example, to improve the usability of AES systems in a classroom setting, AES models can be embedded in an intelligent tutoring system that can help an essay writer improve her essay in an iterative fashion through multiple rounds of feedback and revision.

Recommendation #7: We recommend that AES researchers think beyond essay scoring and develop a vision of how AES models can be combined with related AI technologies to develop more complex systems that can broaden the impact of AES research on education, ideally through discussions with other stakeholders such as teachers.

10 Conclusion

While improving performance numbers should remain an important goal for AES, we believe that merely focusing on performance numbers is not healthy for AES research in the long run. Consequently, in this paper, we discussed several issues that we think AES researchers should think about as we move forward and provided seven recommendations based on our observations. We hope these recommendations will help push the field forward.

⁶Care should be taken, however, to ensure that LLMs are not misused to generate essays with malicious content.

Limitations

The views expressed in this opinion piece are necessarily subjective, so it is possible that the reader may disagree with some or all of our recommendations for future research directions. Nevertheless, we have tried to provide explanations behind our recommendations whenever possible. In addition, given the space limitations, we cannot provide a comprehensive overview of the related research in AES, so a reader without the relevant background (e.g., a reader who does not work in on AES) may not be able to appreciate our recommendations. Nevertheless, we have tried our best to make this paper accessible to a general NLP audience.

Ethics Statement

As mentioned in the paper, as the performance numbers of AES models keep increasing, it is easy for researchers, especially junior researchers, to lose track of the big picture. We are not saying that reporting performance improvements is not important, but we do not believe that one's research should be driven solely by performance numbers in the long run. In addition, we noticed that many researchers who work in the area of AES are doing their own things. This, again, is perfectly fine, but what seems to be missing is a shared vision of how the field should proceed. We hope that the issues raised in this paper as well as the views expressed in it would encourage AES researchers to start thinking about how the field should proceed, with the eventual goal of bringing the members of the field together to develop a shared vision. Note that developing a shared vision by no means implies that we are forcing all AES researchers to do the same thing. We view the shared vision as something that could help guide the long-term development of the field, but individual researchers will still have the freedom to (1) continue to work on their own projects; (2) work on projects that contribute to the shared vision; or (3) split their time between these two things.

References

- Yigal Attali and Jill Burstein. 2006. Automated essay scoring with e-rater v. 2.0. *Journal of Technology, Learning, and Assessment*, 4(3).
- Collin F. Baker, Charles J. Fillmore, and John B. Lowe. 1998. *The Berkeley FrameNet project*. In *36th Annual Meeting of the Association for Computational*

Linguistics and 17th International Conference on Computational Linguistics, Volume 1, pages 86–90, Montreal, Quebec, Canada. Association for Computational Linguistics.

- Beata Beigman Klebanov and Michael Flor. 2013. *Word association profiles and their use for automated scoring of essays*. In *Proceedings of the 51st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1148–1158, Sofia, Bulgaria. Association for Computational Linguistics.

- Beata Beigman Klebanov and Nitin Madnani. 2020. *Automated evaluation of writing – 50 years and counting*. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 7796–7810, Online. Association for Computational Linguistics.

- Beata Beigman Klebanov and Nitin Madnani. 2021. *Automated Essay Scoring*. In Graeme Hirst, editor, *Synthesis Lectures in Human Language Technologies*. Morgan & Claypool Publishers.

- Daniel Blanchard, Joel Tetreault, Derrick Higgins, Aoife Cahill, and Martin Chodorow. 2013. TOEFL11: A corpus of non-native English. *ETS Research Report Series*, 2013(2):i–15.

- Ted Briscoe, Ben Medlock, and Øistein Andersen. 2010. *Automated assessment of ESOL free text examinations*. Technical Report UCAM-CL-TR-790, University of Cambridge, Computer Laboratory.

- Hongbo Chen and Ben He. 2013. *Automated essay scoring by maximizing human-machine agreement*. In *Proceedings of the 2013 Conference on Empirical Methods in Natural Language Processing*, pages 1741–1752, Seattle, Washington, USA. Association for Computational Linguistics.

- Yuan Chen and Xia Li. 2023. *PMAES: Prompt-mapping contrastive learning for cross-prompt automated essay scoring*. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1489–1503, Toronto, Canada. Association for Computational Linguistics.

- Madalina Cozma, Andrei Butnaru, and Radu Tudor Ionescu. 2018. Automated essay scoring with string kernels and word embeddings. In *Proc. of ACL*, pages 503–509.

- Ronan Cummins, Meng Zhang, and Ted Briscoe. 2016. *Constrained multi-task learning for automated essay scoring*. In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 789–799, Berlin, Germany. Association for Computational Linguistics.

- Heejin Do, Yunsu Kim, and Gary Geunbae Lee. 2023. *Prompt- and trait relation-aware cross-prompt essay*

874	trait scoring . In <i>Findings of the Association for Computational Linguistics: ACL 2023</i> , pages 1538–1551, Toronto, Canada. Association for Computational Linguistics.	930
875		931
876		
877		
878	Jesse Dodge, Gabriel Ilharco, Roy Schwartz, Ali Farhadi, Hannaneh Hajishirzi, and Noah Smith. 2020. Fine-tuning pretrained language models: Weight initializations, data orders, and early stopping . <i>Preprint</i> , arXiv:2002.06305.	
879		
880		
881		
882		
883	Fei Dong and Yue Zhang. 2016. Automatic features for essay scoring – an empirical study . In <i>Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing</i> , pages 1072–1077, Austin, Texas. Association for Computational Linguistics.	
884		
885		
886		
887		
888	Fei Dong, Yue Zhang, and Jie Yang. 2017. Attention-based recurrent convolutional neural network for automatic essay scoring . In <i>Proceedings of the 21st Conference on Computational Natural Language Learning (CoNLL 2017)</i> , pages 153–162, Vancouver, Canada. Association for Computational Linguistics.	
889		
890		
891		
892		
893		
894	Scott Elliot. 2003. Intellimetric: From here to validity. In <i>Automated Essay Scoring: A Cross-Disciplinary Perspective</i> , pages 71–86. Lawrence Erlbaum Associates, Mahwah, NJ.	
895		
896		
897		
898	Noura Farra, Swapna Somasundaran, and Jill Burstein. 2015. Scoring persuasive essays using opinions and their targets . In <i>Proceedings of the Tenth Workshop on Innovative Use of NLP for Building Educational Applications</i> , pages 64–74, Denver, Colorado. Association for Computational Linguistics.	
899		
900		
901		
902		
903		
904	James Fiacco, David Adamson, and Carolyn Ros. 2023. Towards extracting and understanding the implicit rubrics of transformer based automatic essay scoring models . In <i>Proceedings of the 18th Workshop on Innovative Use of NLP for Building Educational Applications (BEA 2023)</i> , pages 232–241, Toronto, Canada. Association for Computational Linguistics.	
905		
906		
907		
908		
909		
910		
911	Debanjan Ghosh, Aquila Khanam, Yubo Han, and Smaranda Muresan. 2016. Coarse-grained argumentation features for scoring persuasive essays . In <i>Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)</i> , pages 549–554, Berlin, Germany. Association for Computational Linguistics.	
912		
913		
914		
915		
916		
917		
918	Sylviane Granger, Estelle Dagneaux, Fanny Meunier, and Magali Paquot. 2009. <i>International Corpus of Learner English (Version 2)</i> . Presses universitaires de Louvain.	
919		
920		
921		
922	Barbara J. Grosz, Aravind K. Joshi, and Scott Weinstein. 1995. Centering: A framework for modeling the local coherence of discourse . <i>Computational Linguistics</i> , 21(2):203–225.	
923		
924		
925		
926	Yaqiong He, Feng Jiang, Xiaomin Chu, and Peifeng Li. 2022. Automated Chinese essay scoring from multiple traits . In <i>Proceedings of the 29th International Conference on Computational Linguistics</i> , pages 3007–3016, Gyeongju, Republic of Korea. International Committee on Computational Linguistics.	932
927		933
928		934
929		935
	Andrea Horbach, Dirk Scholten-Akoun, Yuning Ding, and Torsten Zesch. 2017. Fine-grained essay scoring of a complex writing task for native speakers . In <i>Proc. of the BEA workshop</i> , pages 357–366.	
	Mohamed A. Hussein, Hesham A. Hassan, and Mohammad Nassef. 2020. A trait-based deep learning automated essay scoring system with adaptive feedback . <i>International Journal of Advanced Computer Science and Applications</i> , 11(5).	936
		937
		938
		939
		940
	Cancan Jin, Ben He, Kai Hui, and Le Sun. 2018. TDNN: A two-stage deep neural network for prompt-independent automated essay scoring . In <i>Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)</i> , pages 1088–1097, Melbourne, Australia. Association for Computational Linguistics.	941
		942
		943
		944
		945
		946
		947
	Zixuan Ke, Winston Carlile, Nishant Gurrapadi, and Vincent Ng. 2018. Learning to give feedback: Modeling attributes affecting argument persuasiveness in student essays . In <i>Proceedings of the Twenty-Seventh International Joint Conference on Artificial Intelligence, IJCAI-18</i> , pages 4130–4136. International Joint Conferences on Artificial Intelligence Organization.	948
		949
		950
		951
		952
		953
		954
		955
	Zixuan Ke, Hrishikesh Inamdar, Hui Lin, and Vincent Ng. 2019. Give me more feedback II: Annotating thesis strength and related attributes in student essays . In <i>Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics</i> , pages 3994–4004, Florence, Italy. Association for Computational Linguistics.	956
		957
		958
		959
		960
		961
		962
	Zixuan Ke and Vincent Ng. 2019. Automated essay scoring: A survey of the state of the art. In <i>Proceedings of the 28th International Joint Conference on Artificial Intelligence</i> , pages 6300–6308.	963
		964
		965
		966
	Beata Beigman Klebanov, Michael Flor, and Binod Gyawali. 2016. Topicality-based indices for essay scoring. In <i>Proc. of the BEA Workshop</i> , pages 63–72.	967
		968
		969
	Rahul Kumar, Sandeep Mathias, Sriparna Saha, and Pushpak Bhattacharyya. 2022. Many hands make light work: Using essay traits to automatically score essays . In <i>Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies</i> , pages 1485–1495, Seattle, United States. Association for Computational Linguistics.	970
		971
		972
		973
		974
		975
		976
		977
	Sanwoo Lee, Yida Cai, Desong Meng, Ziyang Wang, and Yunfang Wu. 2024. Prompting large language models for zero-shot essay scoring via multi-trait specialization . <i>Preprint</i> , arXiv:2404.04941.	978
		979
		980
		981
	Xia Li, Minping Chen, and Jian-Yun Nie. 2020. Sednn: Shared and enhanced deep neural network model for cross-prompt automated essay scoring . <i>Knowledge-Based Systems</i> , 210:106491.	982
		983
		984
		985

986	Annie Louis and Derrick Higgins. 2010. Off-topic essay detection using short prompt texts . In <i>Proc. of the BEA Workshop</i> , pages 92–95.	
987		
988		
989	Watheq Mansour, Salam Albatarni, Sohaila Eltanbouly, and Tamer Elsayed. 2024. Can large language models automatically score proficiency of written essays? Preprint, arXiv:2403.06149.	
990		
991		
992		
993	Jeziel C. Marinho, Rafael T. Anchiêta, and Raimundo S. Moura. 2021. Essay-BR: a Brazilian corpus of essays. In <i>Dataset Showcase Workshop (DSW)</i> , pages 53–64. Porto Alegre: Sociedade Brasileira de Computação.	
994		
995		
996		
997	Sandeep Mathias and Pushpak Bhattacharyya. 2018. ASAP++: Enriching the ASAP automated essay grading dataset with essay attribute scores . In <i>Proceedings of the Eleventh International Conference on Language Resources and Evaluation (LREC 2018)</i> , Miyazaki, Japan. European Language Resources Association (ELRA).	
998		
999		
1000		
1001		
1002		
1003		
1004	Sandeep Mathias and Pushpak Bhattacharyya. 2020. Can neural networks automatically score essay traits? In <i>Proceedings of the Fifteenth Workshop on Innovative Use of NLP for Building Educational Applications</i> , pages 85–91, Seattle, WA, USA → Online. Association for Computational Linguistics.	
1005		
1006		
1007		
1008		
1009		
1010	Danielle S. McNamara, Scott A. Crossley, Rod D. Roscoe, Laura K. Allen, and Jianmin Dai. 2015. A hierarchical classification approach to automated essay scoring. <i>Assessing Writing</i> , 23:35–59.	
1011		
1012		
1013		
1014	Atsushi Mizumoto and Masaki Eguchi. 2023. Exploring the potential of using an ai language model for automated essay scoring . <i>Research Methods in Applied Linguistics</i> , 2(2):100050.	
1015		
1016		
1017		
1018	Huy V. Nguyen and Diane J. Litman. 2018. Argument mining for improving the automated scoring of persuasive essays. In <i>Proc. of AAAI</i> , pages 5892–5899.	
1019		
1020		
1021	Robert Östling, André Smolentzov, Björn Tyrefors Hinnerich, and Erik Höglén. 2013. Automated essay scoring for Swedish . In <i>Proc. of the BEA Workshop</i> , pages 42–47.	
1022		
1023		
1024		
1025	Ellis B. Page. 1967. Grading essays by computer: Progress report. In <i>Proc. of the Onvitational Conference on Testing Problems</i> .	
1026		
1027		
1028	Isaac Persing, Alan Davis, and Vincent Ng. 2010. Modeling organization in student essays . In <i>Proceedings of the 2010 Conference on Empirical Methods in Natural Language Processing</i> , pages 229–239, Cambridge, MA. Association for Computational Linguistics.	
1029		
1030		
1031		
1032		
1033		
1034	Isaac Persing and Vincent Ng. 2013. Modeling thesis clarity in student essays . In <i>Proceedings of the 51st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)</i> , pages 260–269, Sofia, Bulgaria. Association for Computational Linguistics.	
1035		
1036		
1037		
1038		
1039		
	Isaac Persing and Vincent Ng. 2014. Modeling prompt adherence in student essays . In <i>Proceedings of the 52nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)</i> , pages 1534–1543, Baltimore, Maryland. Association for Computational Linguistics.	1040
		1041
		1042
		1043
		1044
		1045
	Isaac Persing and Vincent Ng. 2015. Modeling argument strength in student essays . In <i>Proceedings of the 53rd Annual Meeting of the Association for Computational Linguistics and the 7th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)</i> , pages 543–552, Beijing, China. Association for Computational Linguistics.	1046
		1047
		1048
		1049
		1050
		1051
		1052
	Peter Phandi, Kian Ming A Chai, and Hwee Tou Ng. 2015. Flexible domain adaptation for automated essay scoring using correlated linear regression. In <i>Proc. of EMNLP</i> , pages 431–439.	1053
		1054
		1055
		1056
	Robert Ridley, Liang He, Xin-Yu Dai, Shujian Huang, and Jiajun Chen. 2021. Automated cross-prompt scoring of essay traits. In <i>Proceedings of AAAI</i> , pages 13745–13753.	1057
		1058
		1059
		1060
	Robert Ridley, Liang He, Xinyu Dai, Shujian Huang, and Jiajun Chen. 2020. Prompt agnostic essay scorer: A domain generalization approach to cross-prompt automated essay scoring. <i>ArXiv</i> , abs/2008.01441.	1061
		1062
		1063
		1064
	Mark D. Shermis, Jill Burstein, Derrick Higgins, and Klaus Zechner. 2010. Automated essay scoring: Writing assessment and instruction. In <i>International Encyclopedia of Education</i> , 3rd edition. Elsevier, Oxford, UK.	1065
		1066
		1067
		1068
		1069
	Chufan Shi, Haoran Yang, Deng Cai, Zhisong Zhang, Yifan Wang, Yujiu Yang, and Wai Lam. 2024. A thorough examination of decoding methods in the era of llms . Preprint, arXiv:2402.06925.	1070
		1071
		1072
		1073
	Christian Stab and Iryna Gurevych. 2014. Annotating argument components and relations in persuasive essays . In <i>Proc. of COLING</i> , pages 1501–1510.	1074
		1075
		1076
	Kaveh Taghipour and Hwee Tou Ng. 2016. A neural approach to automated essay scoring . In <i>Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing</i> , pages 1882–1891, Austin, Texas. Association for Computational Linguistics.	1077
		1078
		1079
		1080
		1081
	Masaki Uto, Yikuan Xie, and Maomi Ueno. 2020. Neural automated essay scoring incorporating hand-crafted features . In <i>Proceedings of the 28th International Conference on Computational Linguistics</i> , pages 6077–6088, Barcelona, Spain (Online). International Committee on Computational Linguistics.	1082
		1083
		1084
		1085
		1086
		1087
	Sowmya Vajjala. 2018. Automated assessment of non-native learner essays: Investigating the role of linguistic features. <i>International Journal of Artificial Intelligence in Education</i> , 28(1):79–105.	1088
		1089
		1090
		1091
	Henning Wachsmuth, Khalid Al-Khatib, and Benno Stein. 2016. Using argument mining to assess the argumentation quality of essays . In <i>Proceedings of</i>	1092
		1093
		1094

COLING 2016, the 26th International Conference on Computational Linguistics: Technical Papers, pages 1680–1691, Osaka, Japan. The COLING 2016 Organizing Committee.

Yongjie Wang, Chuang Wang, Ruobing Li, and Hui Lin. 2022. [On the use of bert for automated essay scoring: Joint learning of multi-scale essay representation](#). In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 3416–3425, Seattle, United States. Association for Computational Linguistics.

Changrong Xiao, Wenxing Ma, Sean Xin Xu, Kunpeng Zhang, Yufang Wang, and Qi Fu. 2024. [From automation to augmentation: Large language models elevating essay scoring landscape](#). *Preprint*, arXiv:2401.06431.

Chaojun Xiao, Xueyu Hua, Zhiyuan Liu, Cunchao Tu, and Maosong Sun. 2021. Lawformer: A pre-trained language model for Chinese legal long documents. *AI Open*, 2:79–84.

Ruosong Yang, Jiannong Cao, Zhiyuan Wen, Youzheng Wu, and Xiaodong He. 2020. [Enhancing automated essay scoring performance via fine-tuning pre-trained language models with combination of regression and ranking](#). In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 1560–1569, Online. Association for Computational Linguistics.

Helen Yannakoudakis and Ted Briscoe. 2012. [Modeling coherence in ESOL learner texts](#). In *Proceedings of the Seventh Workshop on Building Educational Applications Using NLP*, pages 33–43, Montréal, Canada. Association for Computational Linguistics.

Helen Yannakoudakis, Ted Briscoe, and Ben Medlock. 2011. [A new dataset and method for automatically grading ESOL texts](#). In *Proceedings of the 49th Annual Meeting of the Association for Computational Linguistics: Human Language Technologies*, pages 180–189, Portland, Oregon, USA. Association for Computational Linguistics.

Torsten Zesch, Michael Wojatzki, and Dirk Scholten-Akoun. 2015. [Task-independent features for automated essay grading](#). In *Proceedings of the Tenth Workshop on Innovative Use of NLP for Building Educational Applications*, pages 224–232, Denver, Colorado. Association for Computational Linguistics.